

ANTYABHA RAHMAN

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EDUCATION

University of New South Wales

Bachelor of Science in Computer Science **WAM: DN**

Sep. 2023 – Present

Sydney, NSW

The University of British Columbia

Exchange Program

August 2025 – Dec 2025

Vancouver, Canada

EXPERIENCE

Children's Cancer Institute

Capstone Project

Feb 2026 – May 2026

Sydney, NSW

- Migrated an internal gene-scoring pipeline from pandas/MySQL to PySpark and Spark SQL on Azure Databricks, reducing end-to-end runtime from **~4 hours to less than 2 mins**.
- Refactored 4 core scoring functions and ~30 helpers into a modular OOP architecture on Spark, preserving all API contracts to ensure **zero disruption to dependent services**.
- Re-implemented a rule-based gene scoring system as an Explainable Boosting Machine (InterpretML) service, **improving accuracy and recall from ~68% to 90%** and integrating with a clinician-facing dashboard for per-case feature importance inspection.

Algoverse

AI Research Intern |  [GitHub](#)

May 2025 – January 2026

Remote

- First-authored a peer-reviewed paper **accepted at AAAI 2026**, probing internal representations across 4 LLM families with **83-98% accuracy**.
- Architected a large-scale inference pipeline using vLLM on an NVIDIA GH200 GPU, processing **15,000+ responses** to benchmark AI reasoning consistency.
- Identified a novel **8-point accuracy gap** between RL and fine-tuned models through ablation experiments analyzing 28 transformer layers across DeepSeek and Olmo-3.

PROJECTS

Reliable Transport Protocol | C, POSIX Sockets

- Implemented a reliable transport protocol over **UDP in C**, replicating core TCP mechanisms including **3-way handshake**, ACK/retransmission, and sliding window flow control.

Spatial-Textual Similarity Join | [GitHub](#) | Python, PySpark

- Built a distributed record-matching engine in PySpark satisfying spatial (Euclidean) and textual (Jaccard) thresholds, replacing a naive $O(n^2)$ cross-join with partition-aware filtering.

LEADERSHIP

Arc - UNSW Student Life

eReuse Project Coordinator

August 2024 – December 2024

University of New South Wales

- Managed logistics for 150+ donated devices under tight storage constraints, fulfilling **95%** of repair requests by aligning hardware specs with coursework requirements.
- Increased volunteer engagement by **60%** by introducing structured Pair-Repair sessions and community-building events.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, Bash

Tools/Platforms: Git, GitHub, Google Cloud Platform, Databricks, AWS, Linux/Unix, Docker, Kubernetes, Pytest

Frameworks/Technologies: PyTorch, vLLM, PostgreSQL, Apache Spark, Apache Iceberg, Huggingface